DRAFT FOR ORGANIZER REVIEW

AI-Assisted Tool Building for Wall & Ceiling BIM

A two-session workshop for the SWACCA BIM Summit 2026. It shows BIM and VDC staff how to use AI coding agents to build Revit tools, and sends them home with a package they can keep working from.

EVENT	BIM Summit 2026
SLOTS	Ed. Sessions 1 & 2
DATE	Thu 10/29/2026
REV	A · 10/15/2026

At a glance

SESSION 1	SESSION 2	AUDIENCE	TAKEAWAY
Thu 8:30–9:30	**Thu 11:00–12:00**	**Mixed levels**	**Repo + process guide**
Right after the keynote and the Challenge topic vote	After team formation, before lunch and build time	VDC staff, detailers, estimators, field leads, a few in-house scripters	A working Revit tool and the documented steps we used to build it

Two hours is not enough to teach this subject. The sessions are the on-ramp. The **takeaway package is the real product**: attendees bring it back to the office and keep going.

What attendees should do on Monday

THE TARGET WORKFLOW

The goal is a repeatable way to start, not a finished plugin. Back at their desk, an attendee should be able to:

1	**Think the idea through with an LLM** and plan it before writing any code.
2	**Research what already exists** (Dynamo graphs, pyRevit tools, open-source add-ins, vendor features) to reuse, adapt, or learn from.
3	**Gather the APIs and SDKs** the idea needs, starting with the Revit API.
4	**Set up a repository** for the project.
5	**Install and configure Claude Code or Codex.**
6	**Write the project memory first** (CLAUDE.md or AGENTS.md), so the agent keeps the rules and context between sessions.

Levels of engagement

ONE ROOM, FOUR DEPTHS

The session opens by polling the room on experience, and the pacing follows the answers. Every level gets something, and no one gets left behind.

LEVEL	WHO	WHAT THEY DO	WORKSHOP WEIGHT
L0 Spectator	Managers, field leads, estimators	Ask the Revit model questions in an AI chat app. Never see code.	A short demo of what's possible
L1 Scripter	Dynamo / pyRevit-curious BIM staff	Have an LLM write, explain, and fix quick scripts.	Real time on stage
L2 Agent builder	BIM staff ready to build tools	Use Claude Code or Codex in a repo to extend a C# add-in.	The core of the takeaway package
L3 Systems builder	In-house developers	Build MCP tools, use a coder/reviewer split, support multiple Revit versions, write tests.	Shown on stage, covered in depth in the package

The reference tool

BUILT BEFORE THE SUMMIT

`WORKING NAME · WALL & CEILING MODEL INVENTORY`

The week before the summit, we will build a small, **read-only** Revit tool with an AI agent and document every step. At the summit we walk through how it was built rather than live-coding it. Attendees take home the repo and the process guide.

What it does

- Reads a model and reports wall types, layers, fire and STC ratings, heights, lengths, and net area after openings
- Reports ceiling types and heights
- Groups everything by level and exports to Excel or CSV
- Lets an AI agent ask the model these questions directly through **MCP** (Model Context Protocol)

Why this tool

- Remapping generic architect wall types by hand is the most-cited pain point in our research
- Nothing free and current extracts trade-scoped quantities from Revit
- Read-only, so it is safe to run on live projects
- **Runs on Revit 2022–2027.** Autodesk's official Revit MCP server needs Revit 2027.2 or later, which most firms aren't on yet

The same tool appears at every level: L0 asks it questions, L1 sees it as a short pyRevit script, L2 extends the add-in, and L3 adds a new MCP tool.

This builds on what SWACCA members said at the 2025 summit, where custom plug-in development was a session topic and the first takeaway was:

> *"Having software development skills in-house can be a major differentiator."*

Three ways to shape the two hours

FOR DISCUSSION

We haven't picked one. The right shape depends on the room and on how the Summit Challenge runs, so we'd like your input. Bar widths are proportional to minutes.

Everyone | L0 Spectator | L1 Scripter | L2 Agent builder | L3 Systems | Challenge tie-in

A Climb the ladder

Session 1 covers the lower levels and Session 2 the upper ones. Simple and linear.

S1 **What AI can do, and quick scripting**

8 min	10 min	12 min	25 min	5
Room poll + why now	Ask the model questions	How LLMs and agents work	AI-written pyRevit scripts, live	Q&A

S2 **Building real tools with agents**

15 min	15 min	15 min	10 min	5
Idea → research → repo	CLAUDE.md + the add-in	MCP + coder/reviewer	Package tour	Q&A

STRENGTH

Each session has a clear audience.

RISK

Each half of the room waits through the other half's hour.

B The story of the build

Session 1 tells how we built the tool in a week. Session 2 shows it at every level and hands it off.

S1 **How we built a Revit tool with AI in a week**

7 min	8 min	10 min	12 min	18 min	5
Room poll	Teaser: the finished tool	Thinking it through with an LLM	Research: what exists, which APIs	Repo, CLAUDE.md, the agent builds it	Q&A

S2 **Your level, your next step**

15 min	15 min	10 min	12 min	8 min
Same tool as a pyRevit script	MCP + how review kept it honest	Best practices & pitfalls	Use this on your Challenge today	Package tour + Q&A

STRENGTH

One story everyone can follow, with levels as places to stop.

RISK

Session 1 is mostly watching.

C Show it all, then point it at the Challenge

Session 1 tours every level. Session 2 runs the Monday workflow live on a team's Challenge problem.

S1 **One tool, four levels**

7 min	8 min	12 min	15 min	13 min	5
Room poll	Ask the model	The script	The add-in + CLAUDE.md	MCP + reviewer	Q&A

S2 **Live: from Challenge problem to plan**

10 min	20 min	15 min	10 min	5
Pick a team's problem	Think it through + research with agents	Repo + CLAUDE.md + first build	Best practices + package	Q&A

STRENGTH

Highest energy. Teams apply it within the hour.

RISK

Live work on a problem nobody knows until the 8:00 vote.

Takeaway package

DRAFT CONTENTS

To use

- The reference tool repo: add-in, MCP server, pyRevit version
- Install guide for Revit 2022–2027
- Setup guide for Claude Code and Codex
- CLAUDE.md / AGENTS.md starter templates for Revit projects

To learn from

- The build story: prompts, research, decisions, and dead ends, step by step
- Best practices and the coder/reviewer checklist
- Curated resources: Revit API docs, RevitLookup, pyRevit, add-in templates, MCP servers
- Starter prompts for wall & ceiling tasks

Questions for you

YOUR INPUT SHAPES THE PLAN

1. **The Challenge.** Would you welcome a Session 2 that helps teams apply AI tools to their Challenge problem? Does the 9:45 kickoff still fall between our two sessions?
2. **The room.** Roughly how many attendees, and in what roles? Can we send a two-question pre-survey about Revit version and scripting experience?
3. **Revit versions.** Do you know which versions member firms run? We're assuming about three years behind current.
4. **Laptops.** Will attendees have laptops, and Revit, during the sessions or the build time?
5. **Overlap.** Who gives the keynote, Education Session 3, and the vendor demos? We'd like to complement them, not repeat them.
6. **Distribution.** How should the takeaway package reach attendees: a public repo, the member portal, a USB drive, or email after the event?
7. **Room setup.** Projector, internet access for live demos, and whether we can run Revit on stage.

Draft prepared 10/15/2026 for discussion with the SWACCA organizers. Audience and tool research draws on the attending firms' public sites, SWACCA's Innovation blog and 2025 summit recap, AWCI's Walls & Ceilings magazine, and public Revit developer resources.